Spectral Reconstruction Network From Multispectral Images to Hyperspectral Images: A Multitemporal Case

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***Abstract*— Hyperspectral (HS) satellite data have been widely applied in many fields due to its numerous bands. Along with the advantages of high spectral resolution, HS satellite data are still limited by some disadvantages of high acquisition cost, low revisiting capability, and low spatial resolution. Compared with HS satellites, multispectral (MS) satellites have a large number, large width, strong coverage, and high spatial resolution. Therefore, MS data can be used as the input to the spectral reconstruction (SR) to obtain HS data with high temporal resolution. Better HS data can be obtained by spectral reconstructing with these continuous multitemporal data than with single-temporal data. A multitemporal spectral reconstruction network (MTSRN) is proposed in this article, which is used to reconstruct HS images from multitemporal MS images. The proposed MTSRN comprises multiple single-temporal spectral reconstruction networks (STSRN) for extracting temporal features and a multitemporal fusion network (MTFN). The parallel component alternative (PA) post-processing method enhances the physical plausibility of reconstructed HS data. To demonstrate performance of the proposed method in aspects of multitemporal reconstruction, experiments are conducted on four multitemporal HS and MS satellite datasets. The experimental results prove that the proposed MTSRN obtains better SR results compared with the SR method based on single-temporal information.**

***Index Terms*— Hyperspectral (HS) data, multispectral (MS) data, multitemporal, neural networks, spectral reconstruction (SR), spectral superresolution.**

I. Introduction

SATELLITE hyperspectral (HS) imaging technology captures the electromagnetic spectrum reflection in remote sensing scenes. Due to the characteristics of the large number of spectra, high spectral resolution, and features easily to distinguished, the obtained HS images are widely used in target detection [1], [2], change detection [3], [4], classification [5]–[7], semantic segmentation [8], pest prediction [9], and geological mapping [10]. However, due to the manufacturing technology of sensors and optical components, the cost of HS cameras is very high, resulting in expensive and time-consuming acquisition of HS data. In addition, the instantaneous field of view has to increase to obtain the narrow HS bands, resulting in low spatial resolution. It is challenging to employ remote sensing HS data for continuous and wide-field observations of ground objects due to the restricted number of available HS satellites.

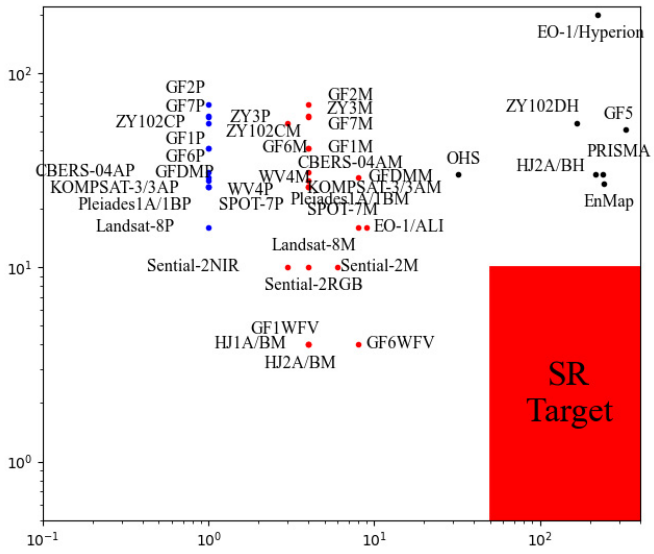


Fig. 1. Contradiction between temporal resolution and spectral resolution of satellite data. The vertical axis is the repeat period and the horizontal axis is the spectral number. Blue: panchromatic data. Red: multispectral data. Black: HS data.

Compared with HS satellite data, multispectral (MS) satellite data have not only the disadvantages of fewer bands and lower spectral resolution, but also the advantages of higher spatial resolution, wider field of view, and fewer mixed components. The wide-field MS satellites designed for wide-field analysis bring the advantages of a short repeat period and high acquisition frequency in the time domain. Repeat periods and spectral numbers of some existing MS satellites and HS satellites are shown in Fig. 1.

To combine the complementary advantages of HS data and MS data, it is a reasonable solution to fuse HS data with MS data or achieve spectral superresolution directly through MS data. However, fusion can only improve the spatial resolution of existing HS data and cannot solve the problem of

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continuous and large-scale observations because of insufficient HS data. The fusion method needs to have HS data in the observation area at the observation time, which cannot solve the problem of insufficient HS data. As a result, to properly benefit from the benefits of MS data's ease of collecting, large quantity, broad short-time space coverage, and high spatial resolution, it is a more suitable approach to reconstruct the MS data straight into HS data.

The short repeat period of MS satellites makes it possible to obtain short-time continuous MS data (e.g., the four-day MS sequence of GF1/GF6 or the five-day MS sequence of Sentinel-2). At present, multitemporal remote sensing data are often used in classification [11], change detection [12], cloud removal [13], intrinsic recovery [14], and other aspects. Due to limited changes of main features in a short-term continuous phase, spectral reconstruction (SR) utilizers information from other continuous time phases to achieve a better reconstruction effect than a single time phase.

Many SR methods have been used to reconstruct RGB/MS data into HS data. However, these methods usually only carry out SR for single data of a single time phase and do not consider the actual situation of data obtained in the remote sensing field. That is, short-time continuous MS remote sensing data can be obtained.

To solve this problem, this article first proposed SR from multitemporal MS data to obtain the corresponding HS data and designed a multitemporal spectral reconstruction network (MTSRN). The MTSRN includes multiple groups of single-temporal spectral reconstruction networks (STSRNs) for extracting temporal features and a multitemporal fusion network (MTFN) for combining multitemporal features.

The main contributions of this article are summarized below.

1) We proposed a SR strategy from multitemporal MS data to obtain the corresponding HS data. To our best knowledge, this is the first time to reconstruct spectral information from multitemporal MS data.
2) A MTSRN combining STSRNs and a MTFN is proposed for SR. The characteristics of the selected phase are extracted by the STSRNs and fed into the MTFN.
3) A fusion network is designed to fuse multisource information. The network realizes multitemporal feature fusion by feature pre-extraction and multiple information fusion blocks.

The remaining part of this article is organized as follows. Section II describes the development of SR methods. We present the framework of the proposed MTSRN in Section III. The results and analysis of the extensive experiment are presented in Section IV. Section V summarizes the conclusions of this work.

II. Related Works

SR methods can be divided into three categories: SR based on matrix mapping, SR methods based on dictionary learning, and SR methods based on deep learning.

A. SR Based on Matrix Mapping

The American scholar Pratt first proposed the SR technique in 1976. The article gave three SR methods: the pseudo-inverse method, the pseudo-inverse method with smoothing constraints, and the Wiener estimation method [15]. These SR methods attempt to complete the reflection from MS data to HS data through a definite spectral reflection matrix. In the following 30 or 40 years, different scholars improved based on these three methods, such as the principal component analysis (PCA) method [16], [17], the improved Wiener estimation method [18], and other methods based on directly solving the mapping matrix or using regression to fit.

The matrix mapping methods represented by the pseudo-inverse method try to find a linear mapping path bound by the datasets shortest mapping.

B. SR Based on Dictionary Learning

American scholars first proposed SR methods based on dictionary learning at ICIP2008 [19].

The SR method based on dictionary learning no longer constructs the direct relation between the MS data and the HS data but obtains the correlation between the two spaces indirectly by constructing the high-dimensional space between the MS dictionary and the HS dictionary. Specifically, the overall mapping relationship is from the MS data space to the MS dictionary space, the HS dictionary space, and finally to the HS data space. Dictionary learning methods based on sparse coding can be divided into two categories according to different ways of constructing dictionaries: 1) obtain the HS dictionary first and then the MS dictionary via the HS dictionary and spectral response function (SRF) [19]–[21] and 2) solve the two dictionaries simultaneously through the constraint relationship between the MS dictionary and the HS dictionary [22]–[24].

The dictionary learning methods attempt to construct a high-dimensional space that is bidirectionally connected to the MS data space and the HS data space via the dictionary space.

C. SR Based on Deep Learning

Deep learning methods are now widely used in remote sensing image processing and SR [25].

In 2003, French scholars proposed using artificial neural networks for SR [26]. NN-based SR methods started to be used [27]. Since 2017, SR methods using convolutional neural network (CNN) and generative adversarial network (GAN) have gradually emerged. Since 2021, some scholars have gradually proposed transformer-based SR methods [28].

SR methods based on GAN can be divided into three types according to the main network structure: SR methods based on conditional generative adversarial network (CGAN) [29]–[33], SR methods based on variational autoencoding generative adversarial network (VAEGAN) [34], [35], and SR methods based on CycleGAN [34], [36].

The methods based on CNN can be divided into three types according to the main network structure.

1) UNet-based reconstruction models [37]–[45]. This type of model will have multiple spatial dimension changes in the network structure, but the input and output spatial dimensions are the same.

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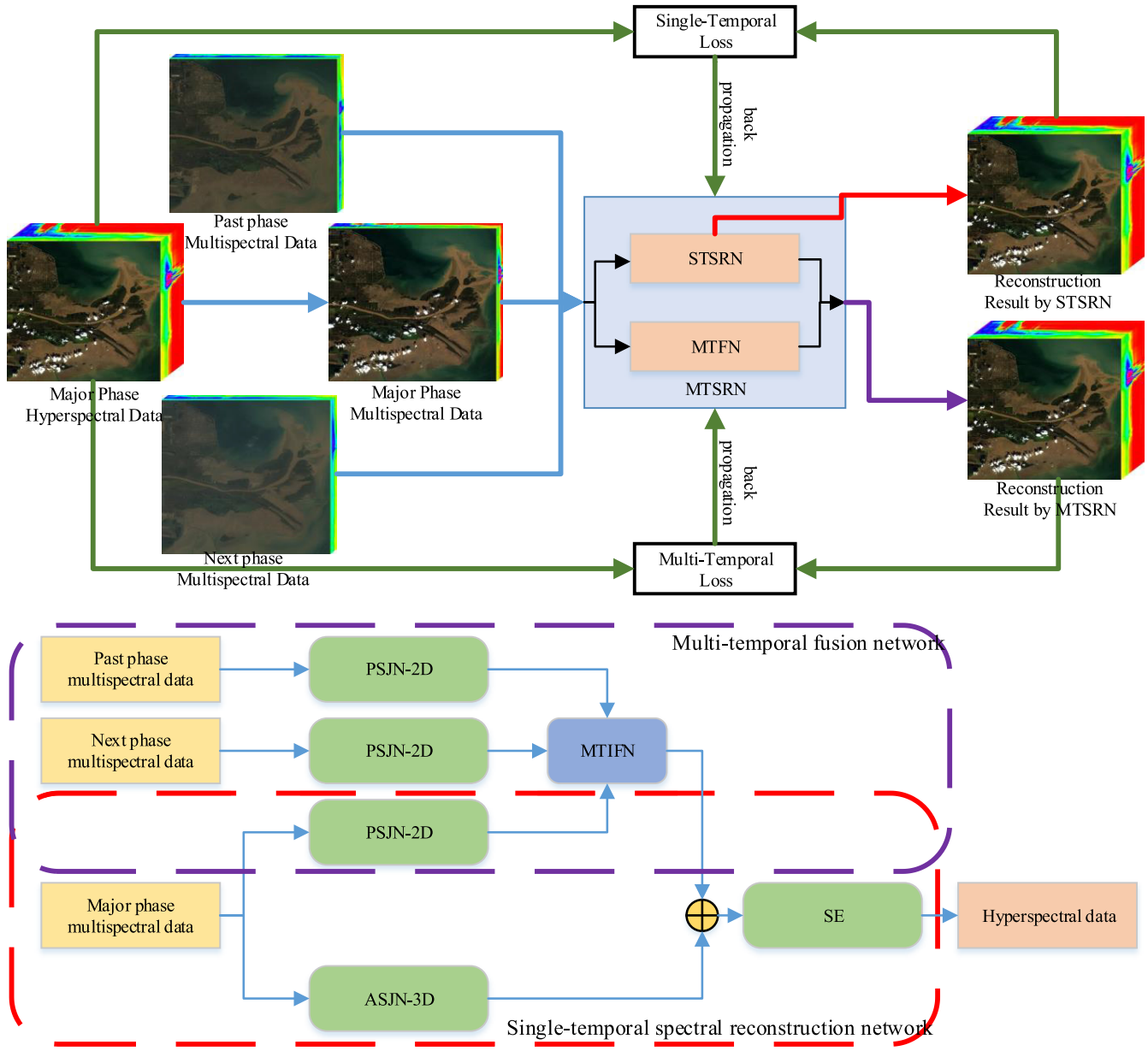


Fig. 2. Workflow of the proposed MTSRN.

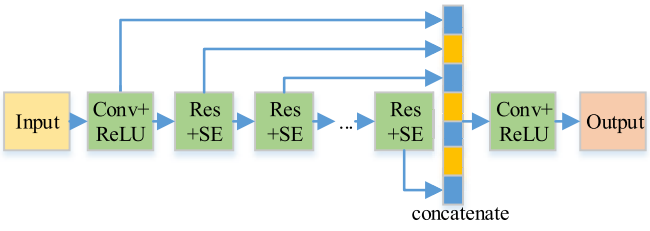


Fig. 3. PSJN2D.

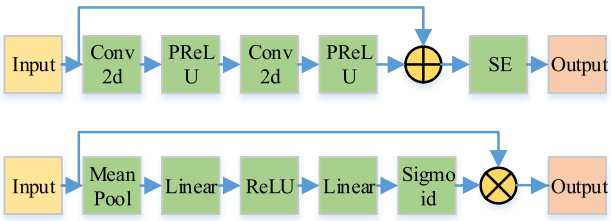


Fig. 4. Res + SE(upper) and SE(down).

2) ResNet-based reconstruction models [46]–[65]. This type of model will not change the spatial dimension during the feature extraction process. It connects the front and back layers of the network through shortcut connections. However, such methods also combine spatial information from neighboring pixels.

3) LeNet5-based reconstruction models [66], [67]. This type of model takes the MS data of a patch area as input and outputs the HS spectrum curve at the center point.

The deep learning methods seek to fit a nonlinear route through a neural network to confine the average error from the reconstructed data to the HS data.

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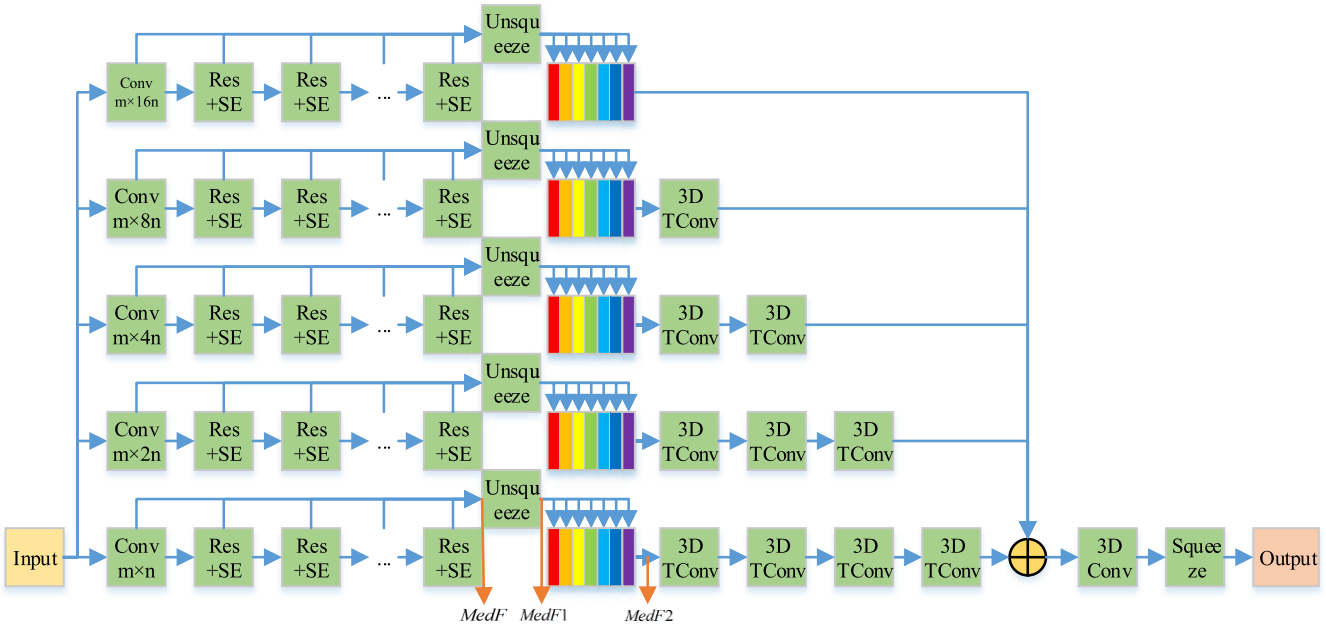


Fig. 5. Network architecture of ASJN3D.

III. Proposed Method

A. Problem Formulation

According to the imaging mechanism of the sensor, the digital-number (DN) value acquired in each band for MS or HS data can be represented as [68]

$$\mathrm{DN}_i = \frac{\int L(\lambda) A \Omega \tau(\lambda) R_i(\lambda) t}{hc/\lambda} \cdot \eta(\lambda) G d\lambda + B_i \tag{1}$$

where λ is the wavelength, $L(\lambda)$ is the continuous spectral radiance, $\tau(\lambda)$ is the spectral transmittance of the lens, $R_i(\lambda)$ is the spectral efficiency, $\eta(\lambda)$ is the quantum efficiency of the detector, A is the area of the entrance pupil, Ω is the solid angle extended by the ground pixel, t is the exposure time, h is the Planck's constant, c is the velocity of light, G is the gain factor of the readout electronics, and B_i is the bias generated by the detector.

Summarize the items related to the sensor's own performance into one item, (1) is equivalent to

$$\mathrm{DN}_i = \int K_i L(\lambda) R_i(\lambda) d\lambda + B_i \tag{2}$$

where $R_i(\lambda)$ is the normalized SRF.

After radiometric calibration, (2) is converted to

$$\overline{L}_i = \int L(\lambda) R_i(\lambda) d\lambda. \tag{3}$$

The formula applies to both of MS and HS data. However, for HS data with a very narrow spectral response curve bandwidth, it can be approximated that the spectral reflectance is consistent within this very narrow spectral response curve bandwidth. Therefore, (3) is equivalent to

$$H_i = \int L_i R_i(\lambda) d\lambda = L_i \int R_i(\lambda) d\lambda = L_i. \tag{4}$$

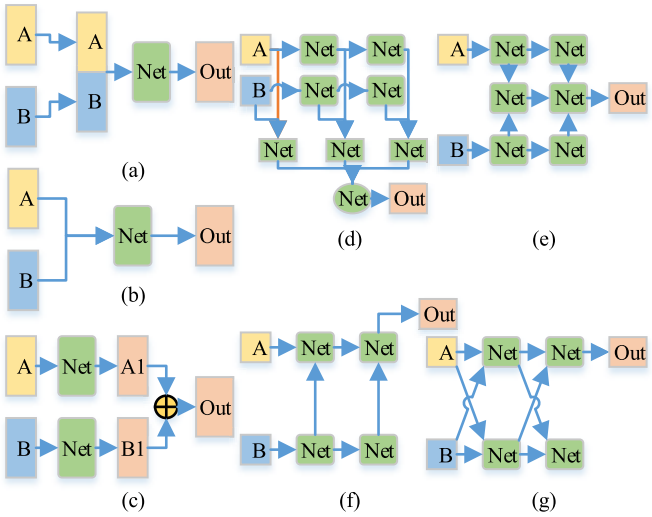


Fig. 6. Data fusion in deep learning. (a) Data-based fusion. (b) Feature-based fusion. (c) Decision-based fusion. (d) Overall feature-based fusion. (e) Layer-by-layer feature-based fusion. (f) Layer-by-layer toward the principal path feature-based fusion. (g) Cross feature-based fusion.

Therefore, MS data can be approximated by HS data and a normalized SRF

$$M_i = \int L(\lambda) R_i(\lambda) d\lambda = \sum \int L(\lambda) R_i(\lambda) = \sum \int H(\lambda) R_i(\lambda). \tag{5}$$

Equation (5) is equivalent to

$$\mathbf{M} = \mathbf{RH} \tag{6}$$

where $\mathbf{H} \in \mathbb{R}^{b_h \times 1}$ is the HS data, $\mathbf{M} \in \mathbb{R}^{b_m \times 1}$ is the MS data, and $\mathbf{R} \in \mathbb{R}^{b_m \times b_h}$ is the spectral response matrix.

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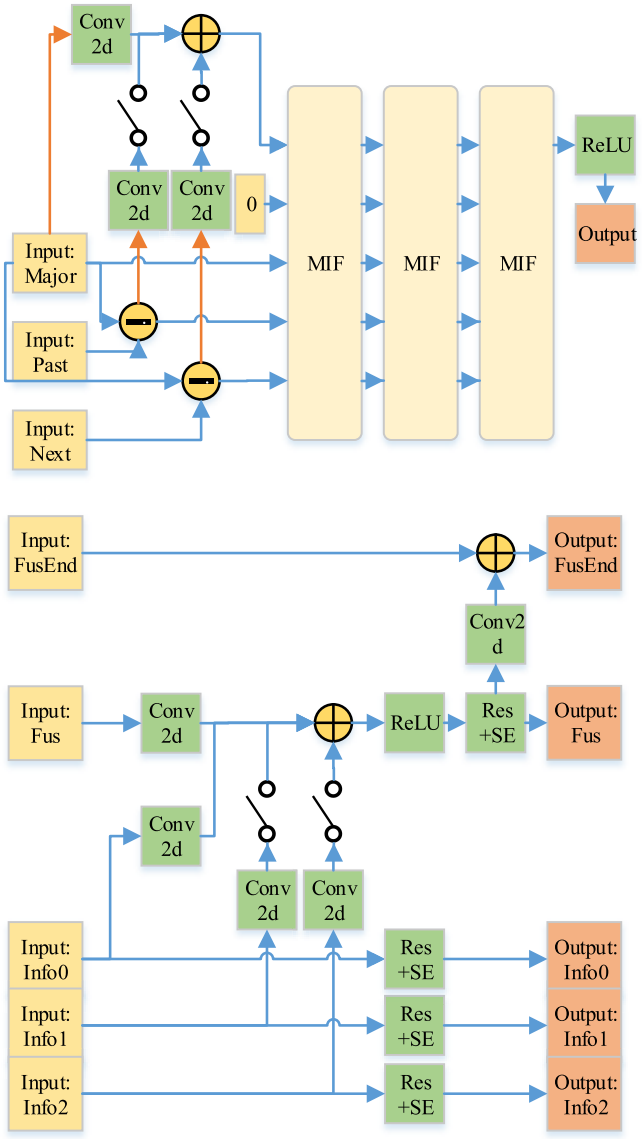


Fig. 7. Network architecture of MTIFN and MIF.

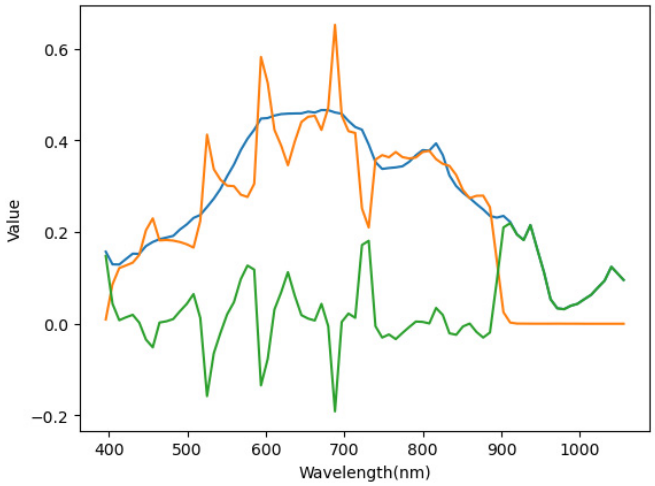


Fig. 8. HS curve of the Yellow River with its parallel component and perpendicular component.

It can now be said that the SR problem can be broken down into the problem of solving **H** with already known **M** and **R**.

For $b_h > b_m$, the linear system is an underdetermined system with infinitely many solutions, and the general solution can be written as

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{Z}u \tag{7}$$

where **Z** is the basis of **R** in null space, namely $\mathbf{RZ} = 0$, $\mathbf{H}_0$ is any particular solution.

Although (6) cannot be solved directly by the matrix method, the mapping from MS data to HS data can be constructed directly by the deep learning method, as shown in the following equation:

$$f_S(\boldsymbol{M}; \theta_S) \rightarrow \boldsymbol{H}. \tag{8}$$

The mapping can be solved by minimizing the objective function

$$\arg\min_{\theta_S} L_1(f_S(\boldsymbol{M}; \theta_S), \boldsymbol{H}) \tag{9}$$

where f_S is the mapping function, θ_S is the network parameter, and L_1 is the loss function.

Considering multitemporal data, the mapping function is

$$f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M) \rightarrow \boldsymbol{H} \tag{10}$$

and the constraint is

$$\arg\min_{\theta_M} L_1(f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M), \boldsymbol{H}) \tag{11}$$

where f_M is the mapping function of multitemporal data, $\boldsymbol{M}$ is the major phase data, $\boldsymbol{M}_1$ is the past phase data, $\boldsymbol{M}_2$ is the next phase data, and θ_M is the network parameter.

It is possible to obtain (12) and (13) using (6), (8), and (10)

$$\boldsymbol{R} f_S(\boldsymbol{M}; \theta_S) \rightarrow \boldsymbol{RH} = \boldsymbol{M} \tag{12}$$

$$\boldsymbol{R} f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M) \rightarrow \boldsymbol{RH} = \boldsymbol{M}. \tag{13}$$

As a result, we may add another constraint to minimize the objective function

$$\arg\min_{\theta_S} L_2(\boldsymbol{R} f_S(\boldsymbol{M}; \theta_S), \boldsymbol{M}) \tag{14}$$

$$\arg\min_{\theta_M} L_2(\boldsymbol{R} f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M), \boldsymbol{M}). \tag{15}$$

The following equation can be obtained by combining (9) and (14):

$$\arg\min_{\theta_S} (L_1(f_S(\boldsymbol{M}; \theta_S), \boldsymbol{H}) + \lambda L_2(\boldsymbol{R} f_S(\boldsymbol{M}; \theta_S), \boldsymbol{M})). \tag{16}$$

The following equation can be obtained by combining (11) and (15):

$$\arg\min_{\theta_M} (L_1(f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M), \boldsymbol{H}) + \lambda L_2(R f_M(\boldsymbol{M}, \boldsymbol{M}_1, \boldsymbol{M}_2; \theta_M), \boldsymbol{M})). \tag{17}$$

Thus, two general objective functions can be constructed to train the network model.

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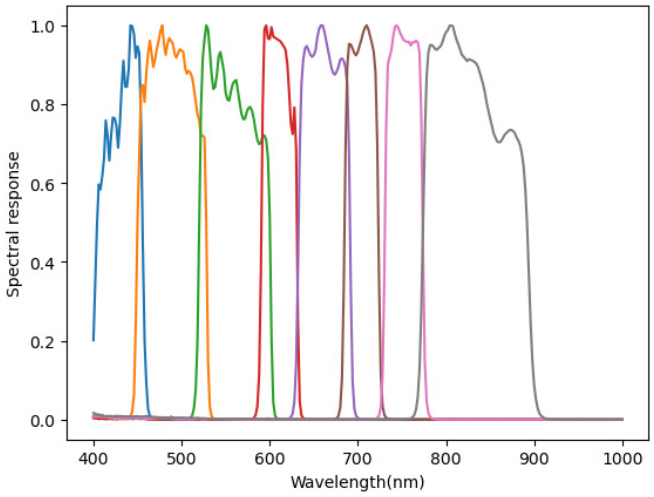


Fig. 9. SRF of GF6WFV.

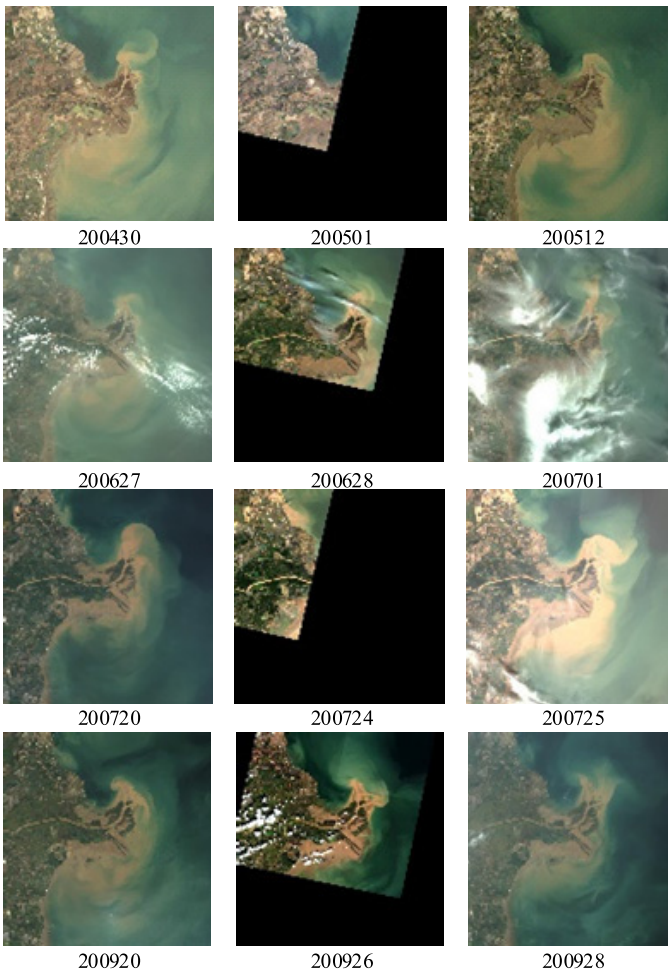


Fig. 10. Twelve scene HS/MS data used in experiment.

B. Workflow and Network Architecture

The overall workflow of this article is shown in Fig. 2. The upper half of the figure describes the workflow of the proposed approach, and the lower half shows the overall structure of the network designed. Three continuous MS data are regarded as a group and are input to MTSRN (where the MS data of the major phase is sampled from the real HS data under the SRF), and two reconstructed HS data output from STSRN and MTSRN are obtained. Backpropagation is used to train the network, resulting in an effective MTSRN.

One part of our MTSRN is called the STSRN, and the other part is called the MTFN. These two parts work together to make the MTSRN.

STSRN consists of three parts: a 2-D spatial feature extraction network (named PSJN2D), a 3-D addative spatial–spectral feature construction network (named ASJN3D), and a channel attention network (named SE).

MTFN consists of two parts: a PSJN2D used for multitemporal feature extraction and a multitemporal information fusion network (MTIFN) for fusing information from different phases.

The mapping relationship of the entire network can be expressed as follows:

$$f_{\mathrm{SE}}(f_{\mathrm{PSJN2D}}(\boldsymbol{M}) + f_{\mathrm{ASJN3D}}(\boldsymbol{M})) \rightarrow \boldsymbol{H} \tag{18}$$

$$\boldsymbol{H}_{\mathrm{MTFN}} = f_{\mathrm{MTIFN}}(f_{\mathrm{PSJN2D}}(\boldsymbol{M}),\, f_{\mathrm{PSJN2D1}}(\boldsymbol{M}_1),\, f_{\mathrm{PSJN2D2}}(\boldsymbol{M}_2))$$
$$f_{\mathrm{SE}}(\boldsymbol{H}_{\mathrm{MTFN}} + f_{\mathrm{ASJN3D}}(\boldsymbol{M})) \rightarrow \boldsymbol{H}. \tag{19}$$

C. PSJN2D

Fig. 3 shows that PSJN2D comprises three parts: a dimensionality-ascending layer, a few feature-processing layers, and a dimensionality-decreasing layer. The dimensionality-ascending layer and dimensionality-decreasing layer consist of a 2-D convolution layer and an activation layer. Multiple residual channel attention modules (Res + SE) and a feature connection layer comprise the feature-extraction layer.

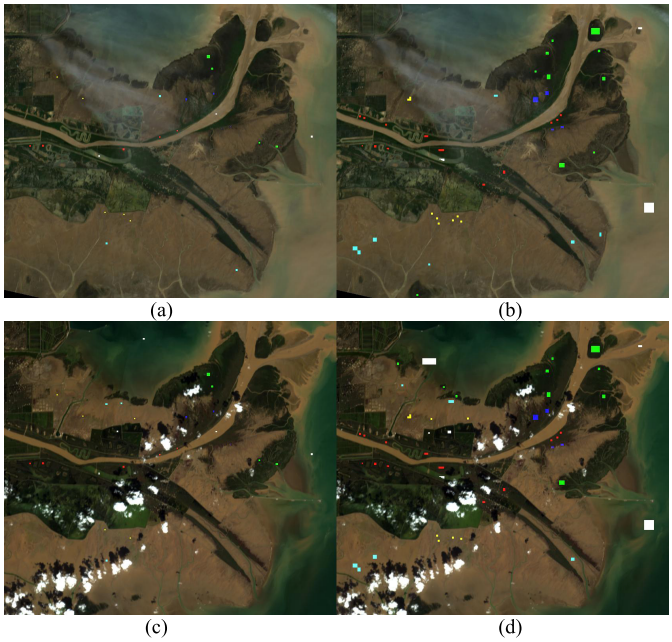


Fig. 11. Training and test truth maps used in 0628 and 0926. (a) Training dataset of 0628. (b) Test dataset of 0628. (c) Training dataset of 0926. (d) Test dataset of 0926.

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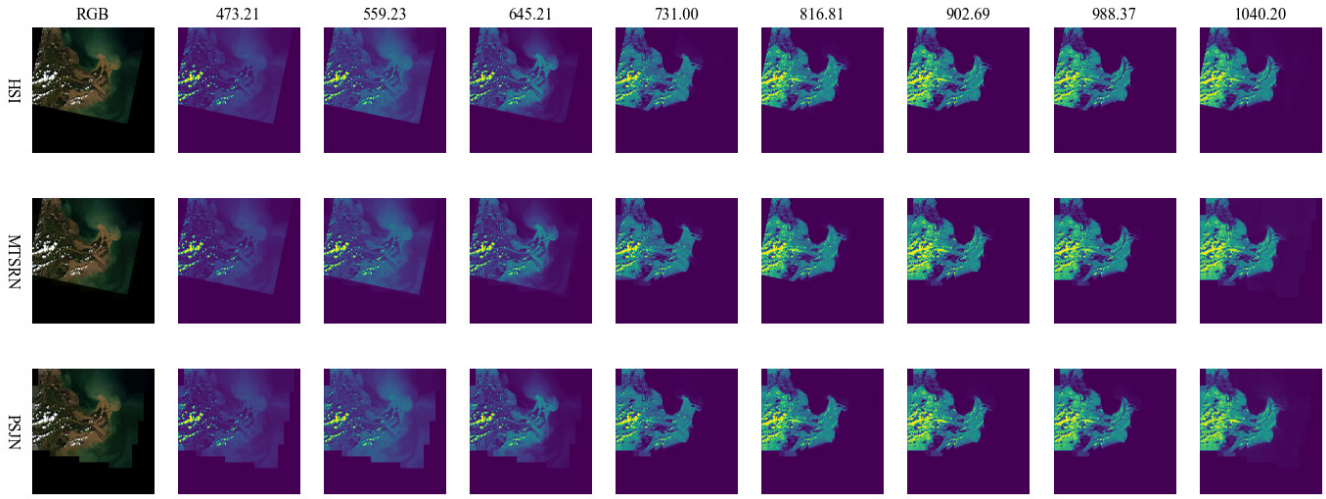


Fig. 12. Schematic of the real HS data and the HS data reconstructed by MTSRN and PSJN in each band in experiment 1.

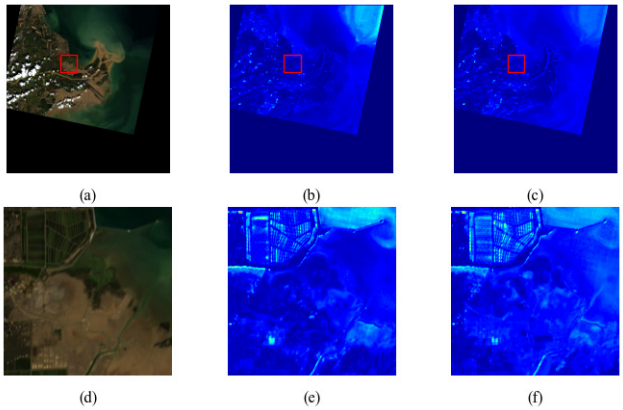


Fig. 13. SAM images for the comparable techniques in experiment 1. (a) RGB composite image. (b) SAM image of SR result by MTSRN. (c) SAM image of SR result by PSJN. (d) RGB composite image in the red box area. (e) SAM image of MTSRN's SR result in the red box area. (f) SAM image of PSJN's SR result in the red box area.

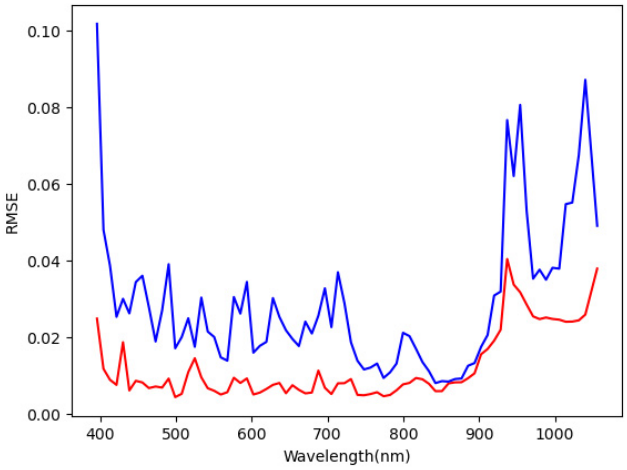


Fig. 14. Comparison of RMSE in each band. Red: MTSRN, Blue: PSJN.

As shown in Fig. 4, the Res + SE module consists of a residual block (Res) and a channel attention block (Squeeze-and-Excitation, SE). Two 2-D convolution layers and two PRelu layers alternate in the residual block. The SE block comprises a global average pooling layer, two fully connected layers, a Relu layer, and a sigmoid layer.

D. ASJN3D

There is a commonly used network structure in spatial super-resolution: $I_{SR} = I \uparrow +\text{Res}$. Inspired by this, a 3-D additive spatial–spectral feature reconstruction network (ASJN3D) was designed, as shown in Fig. 5.

ASJN3D consists of several sub-blocks of specified dimensions and the last addition and dimension-reduction layer. Each sub-block has a channel dimension, and the channel dimension of each sub-block is $\{n, n \times 2, n \times 2^2, \ldots, n \times 2^{k-1}\}$. Where n is the smallest channel dimension in the sub-block, called the basic channel dimension, and k is the number of sub-blocks. Under the conditions of $2 \times h > n \times 2^k > h$ and $n > m$, the values of n and k can be adjusted according to the number of the MS spectrum and HS spectrum.

A sub-block is composed of three parts: a feature-extraction part, a feature-concatenation part, and a transpose-convolution part. The input of all sub-blocks is MS data $M \in \mathbb{R}^{m \times x \times y}$, where x and y are the length and width of the MS data, and m is the number of the MS bands. MS data M is first input into the feature-extraction part, which is composed of a dimension-ascending layer and multiple Res + SE blocks. The structure of the dimension-ascending layer and Res + SE block is consistent with that described in PSJN2D. The dimension ascended through the dimension ascending layer in each sub-block is consistent with the configured channel dimension.

For the pth sub-block, the dimension of the ascending layer and the data output of each Res + SE are consistent with $\text{MedF} \in \mathbb{R}^{(n \times 2^{p-1}) \times x \times y}$. These data are then entered into the feature concatenation part for dimension expansion. At the feature concatenation layer, the feature dimension is added to $\text{MedF1} \in \mathbb{R}^{1 \times (n \times 2^{p-1}) \times x \times y}$ and concatenated into this new dimension. After the concatenation, the dimension of the data is $\text{MedF2} \in \mathbb{R}^{(m+1) \times (n \times 2^{p-1}) \times x \times y}$.

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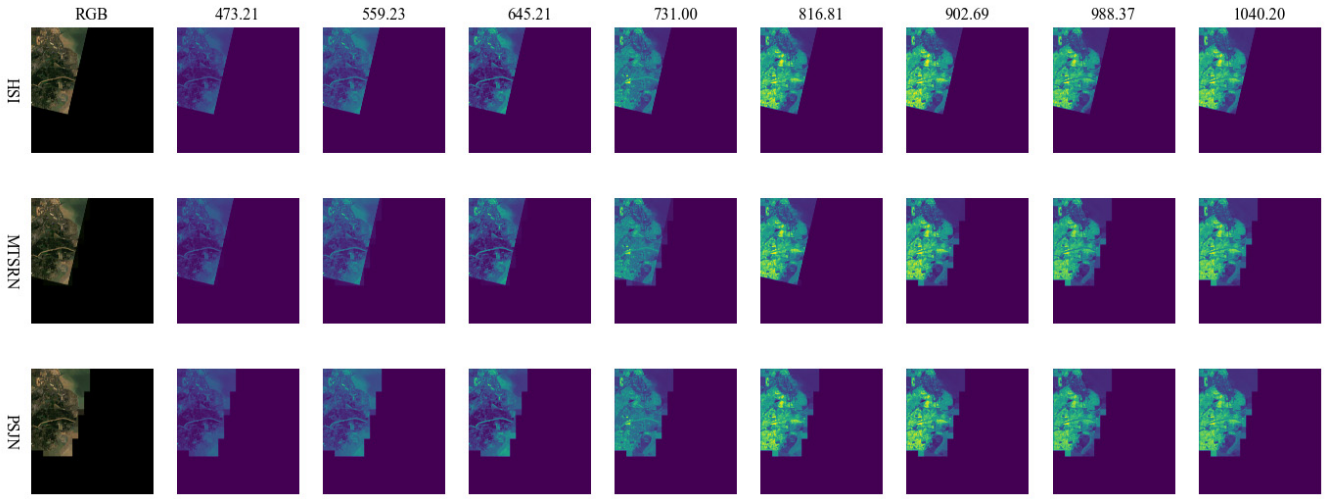


Fig. 15. Schematic of the real HS data and the HS data reconstructed by MTSRN and PSJN in each band in experiment 2.

The concatenated data are input to the deconvolution part. The deconvolution part of the pth sub-block includes $k - p$ deconvolution layers with a step size of 2 (when $p = k$, the sub-block does not contain the deconvolution layer), which is used to promote the data dimension of all sub-blocks to a consistent MedFE $\in \mathbb{R}^{(m+1)\times(n\times 2^{k-1})\times x\times y}$.

After adding the data of the same dimension output by all sub-blocks, the data dimension remains the same as MedFES $\in \mathbb{R}^{(m+1)\times(n\times 2^{k-1})\times x\times y}$. For dimensionality reduction, a 3-D convolution layer with the dimension $[n \times 2^{k-1} + 1 - h, 1, 1]$ of the convolution kernel is used. The dimension of the data after 3-D convolution is $F_{\text{ASJN3DE}} \in \mathbb{R}^{1\times h\times x\times y}$, which becomes $F_{\text{ASJN3D}} \in \mathbb{R}^{h\times x\times y}$ after removing the feature dimension.

E. MTIFN

At present, there are three data fusion methods using deep learning: data-based fusion, feature-based fusion, and decision-based fusion, as shown in Fig. 6(a)–(c), respectively. The feature-based fusion can be roughly divided into four categories according to its network configuration, as shown in Fig. 6(d)–(g).

A MTIFN is designed based on the concept of Fig. 6(e) to make full use of the information brought by multitemporal data. As shown in Fig. 7, the designed network consists of a basic composite block, multiple multi-information fusion blocks (MIF), and an activation layer at the end of the network.

The MTIFN realizes the switching of two modes through switches located in the basic composite block and MIF. The two modes are as follows.

1) When the input contains only one phase, the switch is disconnected and MTIFN processes information from only one phase.
2) When there is more than one phase in the input, the switch is closed, and MTIFN processes information from all phases simultaneously.

The MTIFN first calculates the difference between input features from the major phase and input features from the other two phases (the features here refer to the features obtained through PSJN2D processing) and then adds the two results and input features from the major phase into the basic composite block and the first MIF.

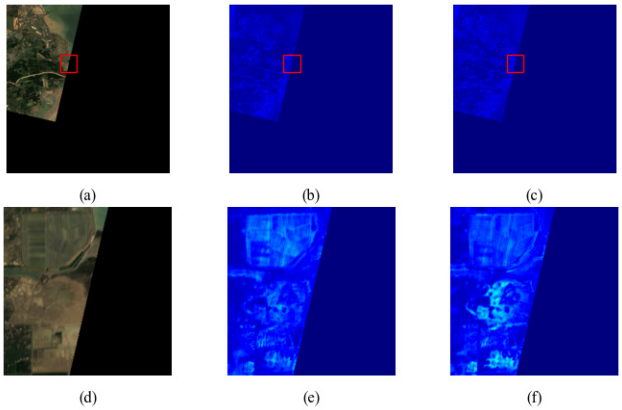


Fig. 16. SAM images for the comparable techniques in experiment 2. (a) RGB composite image. (b) SAM image of SR result by MTSRN. (c) SAM image of SR result by PSJN. (d) RGB composite image in the red box area. (e) SAM image of MTSRN's SR result in the red box area. (f) SAM image of PSJN's SR result in the red box area.

The basic combination block convolutes and sums the inputs from all phases. It should be noted that, except for the branches corresponding to the major phase, other branches are designed with switches to realize the two modes described above.

MIF is a module with five inputs and five outputs, including three information inputs, three information outputs, one fusion information input, one fusion information output, one final fusion information input, and one final fusion information output. The three information branches do not interfere and are output after feature processing by a Res + SE block. The information fusion branch fuses the information from the current branch and the three information branches utilizing convolution, addition, and activation. This information is output after being processed by the Res + SE block features. The final information fusion branch does not interfere with other

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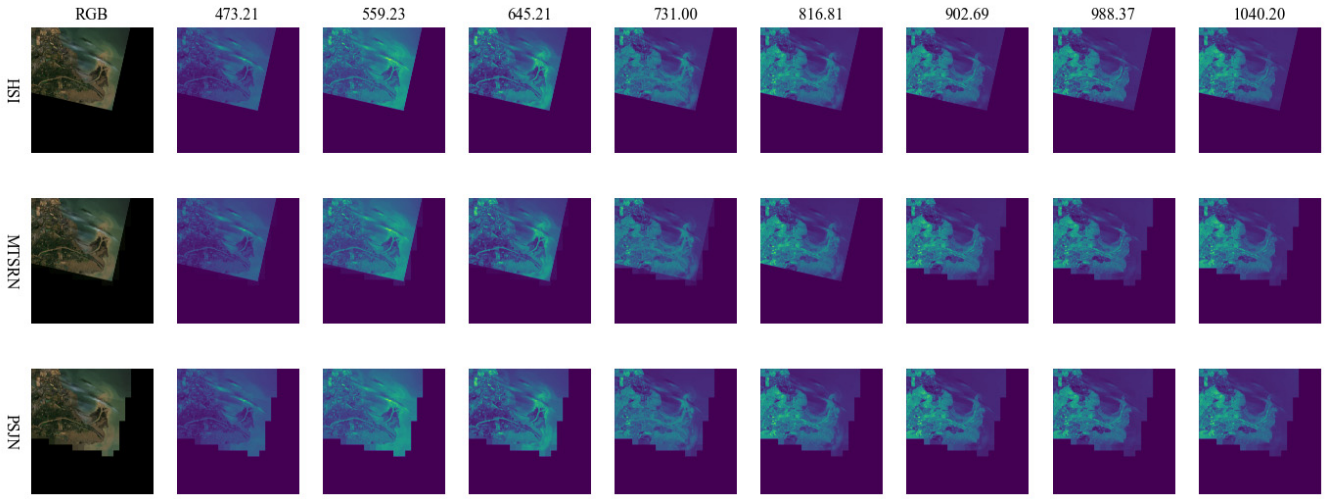


Fig. 17. Schematic of the real HS data and the HS data reconstructed by MTSRN and PSJN in each band in experiment 3.

branches and only combines information from the information fusion branch in the current block with the final information fusion input.

The MIF at the last level only outputs the final fusion information, which is processed by the activation layer as the output of the whole MTIFN.

F. Parallel Component Alternative (PA)

Our previous work [57] ensured the spectral reducibility of reconstructed results through reconstruction loss, but physical plausibility was still not completely realized.

As described in the Problem Formulation section, the solution of (7) consists of two parts, Zu and H_0. Zu can be understood as any vector perpendicular to the hypersurface of the SRF, while H_0 is the sum of any vector perpendicular to the hypersurface of the SRF and a specified vector parallel to the hypersurface of the SRF.

The solution of (6) and (7) in the minimum $L2$ norm yields a unique solution $R^T(RR^T)^{-1}M$, which is the specific vector mentioned above. If this vector is taken as H_0, the HS data can be divided into a parallel component and a perpendicular component. The parallel component can be directly determined by the MS data and SRF, while the perpendicular component cannot be directly determined. Lin and Finlayson [69] tried to fit the perpendicular component through the neural network directly, but this was not quite suitable. As shown in Fig. 8, the blue curve is the HS curve of the Yellow River on September 26, 2020, the orange curve is the parallel component, and the green curve is the perpendicular component. This graph shows that the green curve changes quickly and has a lot of negative values, while the value does not match the MS data, which will make it more difficult for the network to be trained.

We first use the method to obtain SR results through the MTSRN network and then divide them into parallel and perpendicular components. At the same time, the original MS data are used to calculate the corresponding parallel components, which are added with the perpendicular components to obtain the final output of the reconstructed HS data. In this way, the spectral continuity of HS data can be fully utilized in the reconstruction process while ensuring the physical plausibility of the reconstructed spectra. This process is called PA. The final calculation process is shown in the following equation:

$$H_{\mathrm{RC}} = f(M;\theta) - R^T\left(RR^T\right)^{-1}R \times f(M;\theta) + R^T\left(RR^T\right)^{-1}M. \tag{20}$$

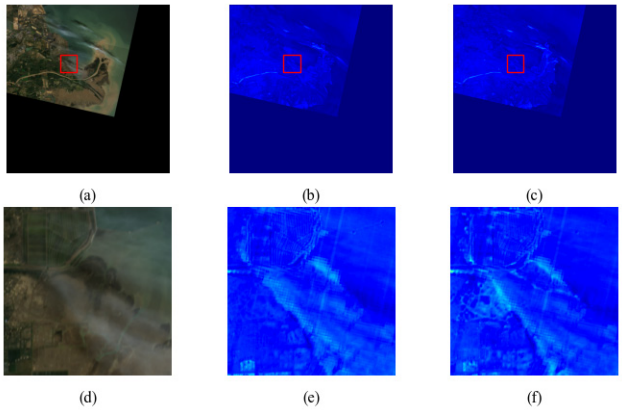


Fig. 18. SAM images for the comparable techniques in experiment 3. (a) RGB composite image. (b) SAM image of SR result by MTSRN. (c) SAM image of SR result by PSJN. (d) RGB composite image in the red box area. (e) SAM image of MTSRN's SR result in the red box area. (f) SAM image of PSJN's SR result in the red box area.

IV. Experiments

A. Dataset Description

Data used in the experiments include HS data from ZY102D and wide-field MS data from GF6.

The ZY102D HS satellite was launched by China in 2019. The satellite will be repeated in 55 days with a spatial resolution of 30 m and a width of 60 km. The HS sensor has 166 bands ranging from 396 to 2501 nm, among which 77 bands ranging from 396 to 1056 nm were selected in this experiment.

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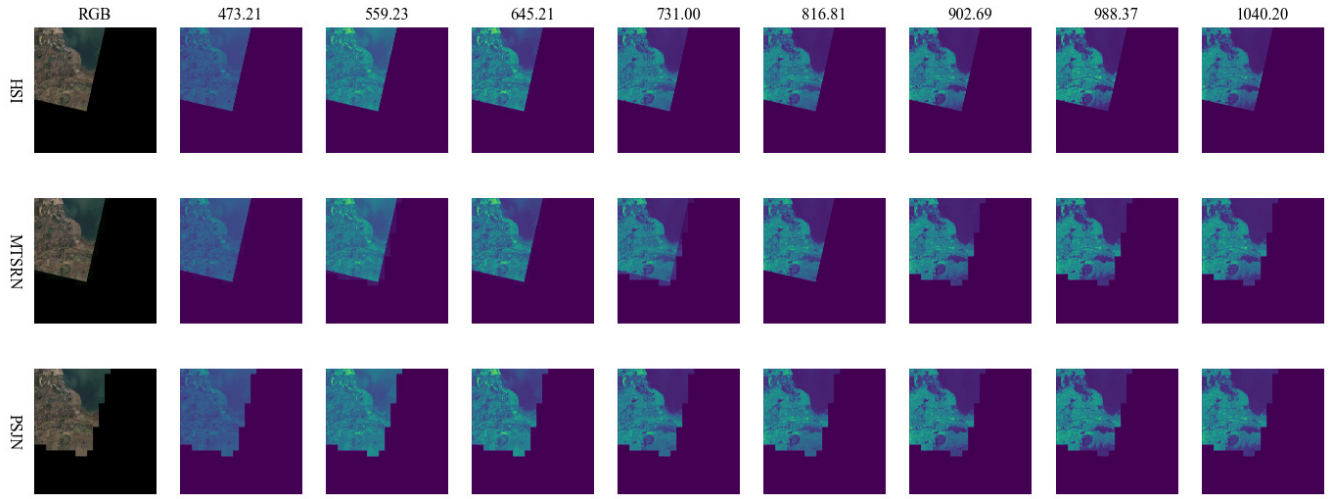


Fig. 19. Schematic of the real HS data and the HS data reconstructed by MTSRN and PSJN in each band in experiment 4.

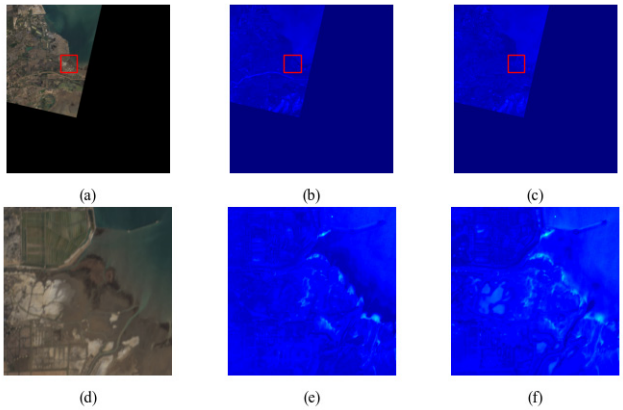


Fig. 20. SAM images for the comparable techniques in experiment 4. (a) RGB composite image. (b) SAM image of SR result by MTSRN. (c) SAM image of SR result by PSJN. (d) RGB composite image in the red box area. (e) SAM image of MTSRN's SR result in the red box area. (f) SAM image of PSJN's SR result in the red box area.

The GF6 MS satellite was launched by China in 2018. GF6 has two imaging modes: narrow-filed panchromatic and multispectral imaging (PMS) and imaging with a wide field of view (WFV). WFV mode imaging data were selected for this experiment. The satellite with WFV mode will be repeated in four days with a spatial resolution of 16 m and a width of 800 km. The SRF of GF6WFV is shown in Fig. 9.

The four scene HS data of the ZY102D HS satellite were obtained on May 1, 2020, June 28, 2020, July 24, 2020, and September 26, 2020, in the Yellow River Delta National Nature Reserve of Dongying, Shandong, China, for this experiment. The GF6WFV data captured in the surrounding time is chosen based on the preceding stage, and the particular time is presented in Table I.

All 12 scenes' data were cropped to the same geographic region. Eight scene images of GF6WFV were registered and resampled to a spatial resolution of 30 m following ZY102D. Fig. 10 depicts the RGB representation of all 12 scenarios.

TABLE I
TWELVE SCENE USED IN EXPERIMENT

Dataset	Major phase	Past phase	Next phase
Dataset 0501	20.05.01 ZY102D HSI	20.04.30 GF6WFV MSI	20.05.12 GF6WFV MSI
Dataset 0628	20.06.28 ZY102D HSI	20.06.27 GF6WFV MSI	20.07.01 GF6WFV MSI
Dataset 0724	20.07.24 ZY102D HSI	20.07.20 GF6WFV MSI	20.07.25 GF6WFV MSI
Dataset 0926	20.09.26 ZY102D HSI	20.09.20 GF6WFV MSI	20.09.28 GF6WFV MSI

By calculating the vector angle between adjacent temporal pixels, the 16 × 16 region with 80% of pixels with vector angles higher than 0.95 is regarded as the region with relevant multitemporal data. It's used for SR in these areas, and the single-temporal mode is used in other areas.

Four sets of experiments were conducted.

1) Datasets 0501, 0628, and 0724 were used as training sets. Dataset 0926 was used as the test set.
2) Datasets 0501, 0628, and 0926 were used as training sets. Dataset 0724 was used as the test set.
3) Datasets 0501, 0724, and 0926 were used as training sets. Dataset 0628 was used as the test set.
4) Datasets 0628, 0724, and 0926 were used as training sets. Dataset 0501 was used as the test set.

HS data 200 628 and 200 926 were classified by the SVM classifier together with the corresponding MS data and SR results. The truth map used for classification contains six categories: water, mudflat, Tamarix, Suaeda, reed, and Spartina. Due to different cloud shielding areas, the training and test sets used for classification are slightly different, as shown in Fig. 11. The number of various samples for training and testing is shown in Tables II and III.

In addition, the 200 928 MS data were spectrally reconstructed and classified, and the distribution of classification labels was consistent with that of 200 926, with the exception of a change in spatial resolution from 30 to 16 m.

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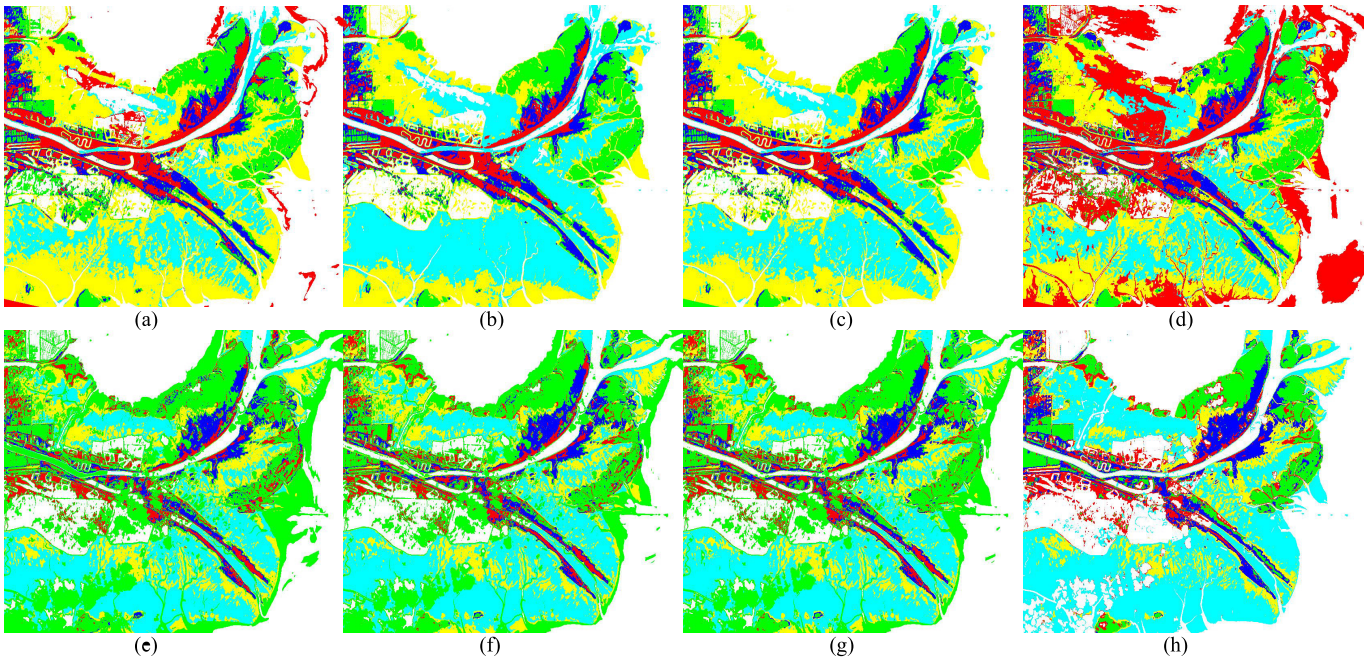


Fig. 21. Whole graph classification result of 0628 and 0926. (a) Classification result of 0628 MSI. (b) Classification result of 0628 SR HSI by PSJN. (c) Classification result of 0628 SR HSI by MTSRN. (d) Classification result of 0628 HSI. (e) Classification result of 0926 MSI. (f) Classification result of 0926 SR HSI by PSJN. (g) Classification result of 0926 SR HSI by MTSRN. (h) Classification result of 0926 HSI.

TABLE II
DETAILS OF THE GROUND TRUTH IN DATASET 0628

0628	Total	Water	Mudflat	Tamarix	Suaeda	Reed	Spartina
Train	460	77	75	41	45	87	135
Test	3381	1029	497	327	225	420	883

TABLE III
DETAILS OF THE GROUND TRUTH IN DATASET 0926

0926	Total	Water	Mudflat	Tamarix	Suaeda	Reed	Spartina
Train	461	88	65	41	45	87	135
Test	3718	1368	495	327	220	420	888

B. Evaluation Metrics

For the HS-MS dataset, the reconstruction quality is measured by the similarity metrics between the reconstructed HS images and real HS images. The specific evaluation metrics are root mean square error (RMSE), mean relative absolute error (MRAE), spectral angle mapper (SAM), and peak-signal-to-noise ratio (PSNR). In addition, to evaluate the physical plausibility of the SR, the RMSE between the ground truth and the re-generated MS images are calculated, which is called PPRMSE. The formulas for RMSE, MRAE, SAM, PSNR, and PPRMSE are as follows:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_c \left(P_{\mathrm{gt}_c} - P_{\mathrm{rec}_c}\right)^2}{N}} \tag{21}$$

$$\mathrm{MRAE} = \frac{\sum_c \frac{|P_{\mathrm{gt}_c} - P_{\mathrm{rec}_c}|}{P_{\mathrm{gt}_c}}}{N} \tag{22}$$

$$\mathrm{SAM} = \cos^{-1}\left(\frac{P_{\mathrm{gt}}^{\mathrm{T}} P_{\mathrm{rec}}}{\left(P_{\mathrm{gt}}^{\mathrm{T}} P_{\mathrm{gt}}\right)^{1/2}\left(P_{\mathrm{rec}}^{\mathrm{T}} P_{\mathrm{rec}}\right)^{1/2}}\right) \tag{23}$$

$$\mathrm{PSNR} = 10 \times \lg\left(\frac{\mathrm{MaxValue}^2}{\mathrm{MSE}}\right) \tag{24}$$

$$\mathrm{PPRMSE} = \sqrt{\frac{\sum_c \left(P_{\mathrm{gt}_c} R - P_{\mathrm{rec}_c} R\right)^2}{N}}. \tag{25}$$

TABLE IV
RECONSTRUCTION PERFORMANCE COMPARISON OF IN TERMS OF FOUR METRICS ON EXPERIMENT 1. THE BEST RESULTS ARE SHOWN IN BOLD

EX1	RMSE	MRAE	SAM	PSNR	PPRMSE
PSJN	0.016507	0.032569	0.087700	39.671	0.004949
MTSRN	0.014923	0.028632	0.090781	40.182	0.004074
MTSRN+PA	**0.014394**	**0.026223**	**0.085837**	**41.636**	**0**

The SVM classifier classified the training samples and test samples to evaluate the classification ability. The evaluation metrics are overall accuracy (OA), average accuracy (AA), kappa coefficient (KAPPA), and accuracy for each class.

C. Similarity-Based Evaluation

Fig. 12 shows the representative bands of the reconstructed HS images obtained using MTSRN based on multitemporal information and PSJN based on single-temporal methods in the 200 926 HS scene (Experiment 1). For the convenience of comparison, the corresponding bands of the real HS images are listed at the top of the figure. We describe eight of the 77 bands: 473.21, 559.23, 645.21, 731.00, 816.81, 902.69,

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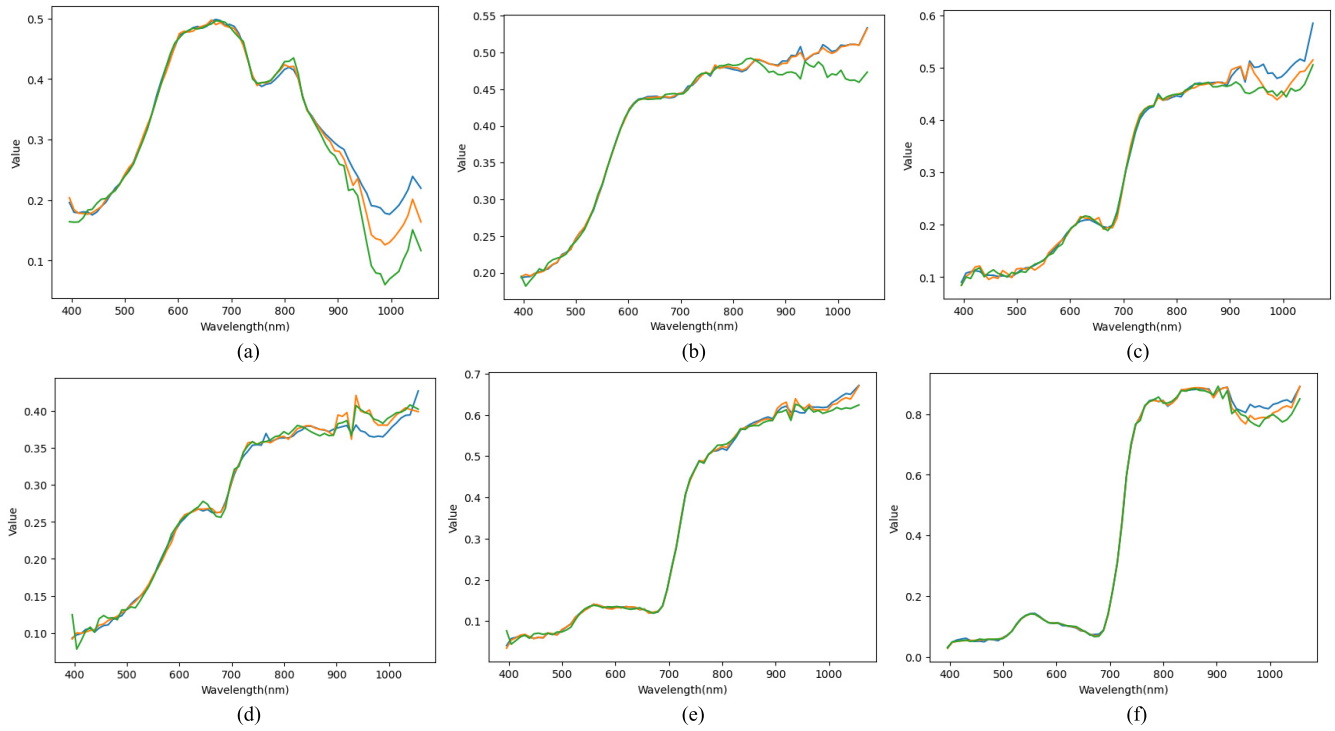


Fig. 22. Spectral reflectance curves for 6 classes in Dataset 0926. Green: Real spectrum, Blue: PSJN, Orange: MTSR. (a) Water. (b) Mudflat. (c) Tamarix. (d) Suaeda. (e) Reed. (f) Spartina.

TABLE V
RECONSTRUCTION PERFORMANCE COMPARISON IN TERMS OF FOUR METRICS ON EXPERIMENT 2. THE BEST RESULTS ARE SHOWN IN BOLD

EX2	RMSE	MRAE	SAM	PSNR	PPRMSE
PSJN	0.022997	0.029868	0.054113	40.308	0.002073
MTSRN	0.020303	0.026991	**0.047567**	41.022	0.001261
MTSRN+PA	**0.020263**	**0.026873**	0.047635	**41.197**	**0**

TABLE VI
RECONSTRUCTION PERFORMANCE COMPARISON IN TERMS OF FOUR METRICS ON EXPERIMENT 3. THE BEST RESULTS ARE SHOWN IN BOLD

EX3	RMSE	MRAE	SAM	PSNR	PPRMSE
PSJN	0.023602	0.031474	0.069077	40.590	0.001210
MTSRN	0.022531	0.030119	0.066020	40.686	0.001673
MTSRN+PA	**0.022487**	**0.029818**	**0.065890**	**40.984**	**0**

TABLE VII
RECONSTRUCTION PERFORMANCE COMPARISON IN TERMS OF FOUR METRICS ON EXPERIMENT 4. THE BEST RESULTS ARE SHOWN IN BOLD

EX4	RMSE	MRAE	SAM	PSNR	PPRMSE
PSJN	0.020148	0.028132	0.061393	41.714	0.000752
MTSRN	0.018736	0.026195	0.056197	42.169	0.001128
MTSRN+PA	**0.018706**	**0.025892**	**0.056066**	**42.506**	**0**

988.37, and 1040.20 nm, which are selected to distinguish the subtle differences in the reconstructed HS spectrum.

The similarity comparison results of 200 926 HS scenes are shown in Table IV. In Experiment 1, MTSRN using multitemporal information significantly improved about 11% in RMSE and MRAE and had a slight decrease of 3% in SAM compared with PSJN using only single-temporal information. The post-processing of the MTSRN output results using PA will further improve RMSE, MRAE, and SAM by about 5%. As for the PPRMSE index, which characterizes the physical plausibility of the reduction, MTSRN has a slight improvement compared with PSJN. The PPRMSE can be reduced to 0 after PA, i.e., the reconstructed HS data can be perfectly projected onto the original MS data by SRF.

The SAM evaluation of the multitemporal and single-temporal reconstruction results is shown in Fig. 13. It can be seen that the multitemporal reconstruction results are better than the single-temporal reconstruction results in the land and offshore areas, while the multitemporal reconstruction results are slightly worse than the single-temporal reconstruction results in the far offshore area. In the detailed analysis of the red box area, we can see that the results obtained using multitemporal SR are better than those obtained by single-temporal SR in the farmland area in the upper left corner, the saline beach area in the middle, and the river area in the lower right corner.

To observe more precisely the differences in reconstruction effects in different bands, the RMSE for each band is presented in Fig. 14, and it can be seen that after combining the multitemporal information, a specific improvement in reconstruction effects is obtained in almost all bands (manifested by lower RMSE values).

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TABLE VIII

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%) AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0628. THE BEST RESULTS ARE SHOWN IN BOLD

0628	OA	AA	KAPPA	Water	Mudflat	Tamarix	Suaeda	Reed	Spartina
MSI	88.85	85.68	85.96	99.90	50.50	85.32	**92.44**	**87.14**	**98.75**
PSJN	91.07	83.84	88.65	**100**	**98.79**	88.07	48.00	69.52	98.64
MTSRN	91.33	86.94	89.03	**100**	85.92	**89.60**	75.56	72.86	97.73
HSI	91.78	89.11	89.62	98.74	76.06	79.20	95.11	86.90	98.64

TABLE IX

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%) AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0926. THE BEST RESULTS ARE SHOWN IN BOLD

0926	OA	AA	KAPPA	Water	Mudflat	Tamarix	Suaeda	Reed	Spartina
MSI	89.03	85.36	85.75	100	100	81.65	89.09	58.10	83.33
PSJN	89.46	87.06	86.34	100	100	84.40	94.09	63.57	80.29
MTSRN	91.20	88.37	88.58	100	100	85.02	91.82	67.14	86.26
HSI	90.59	85.82	87.74	100	100	92.05	81.82	48.81	92.23

TABLE X

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%) AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0928. THE BEST RESULTS ARE SHOWN IN BOLD

0928	OA	AA	KAPPA	Water	Mudflat	Tamarix	Suaeda	Reed	Spartina
MSI	86.54	85.53	82.83	100	84.89	90.17	85.42	71.27	71.43
PSJN	88.92	88.37	85.87	100	85.34	88.28	93.33	90.50	72.78
HSI	91.03	89.16	88.49	100	84.89	89.18	95.83	77.85	87.24

The wavelength display plots of the reconstructed HS data obtained from the remaining three sets of experiments are shown in Figs. 15, 17, and 19.

The similarity comparison results of Experiment 2 are shown in Table V. In Experiment 2, MTSRN has a significant improvement of about 11% in RMSE, MRAE, SAM, and a slight improvement in PPRMSE compared with PSJN. With the addition of the PA posttreatment, there is additional improvement in the four metrics. As in Experiment 1, we give the comparison graph and a detailed enlargement of the SAM values, as shown in Fig. 16. It can be seen that MTSRN obtains a better reconstruction both in the middle and at the bottom of the image.

Table VI displays the result of Experiment 3. In Experiment 3, MTSRN outperforms PSJN by about 4.5 percent in terms of RMSE, MRAE, and SAM. Furthermore, by post-processing the MTSRN output results with PA, the RMSE, MRAE, SAM, and PPRMSE are slightly improved. Fig. 18 depicts a comparison graph and a detailed enlargement of the SAM values.

Experiment 4's experimental results are provided in Table VII. When comparing the reconstruction results of MTSRN to those of PSJN, it can be noticed that the results of MTSRN show a considerable improvement in RMSE, MRAE, and SAM, with an improvement of roughly 7%. Furthermore, using PA to post-process the output of MTSRN, the RMSE, MRAE, SAM, and PPRMSE are marginally improved. We present the comparison graph and a detailed enlargement of the SAM results in Fig. 20.

D. Classification-Based Evaluation

As an advanced application, classification can measure the effect of SR. Classification performance comparisons of different methods in terms of OA, AA, and KAPPA, as well as the accuracy for each class on HSI datasets 0628, 0926, and MSI dataset 0928, are shown in Tables VIII–X. The MS dataset 0928 was spectrally reconstructed using the network trained in Experiment 1. It can be seen that the classification ability (OA, AA, and KAPPA) of the spectrally reconstructed HS images is better than that of the original MS ones in all three sets of classification experiments. The classification ability of the reconstructed results of the multitemporal-based MTSRN is better than that of the reconstructed results of the single-temporal-based PSJN.

The whole graph classification results for the 0628 dataset and the 0926 dataset are shown in Fig. 21. In the 0628 dataset, the HS data obtained by SR are more discriminative for reeds than the original MS data while reducing the misclassification between mudflat and suaeda. In the 0926 dataset, SR helped reduce the misclassification between spartina and water, spartina and reeds in the coastal spartina area. Real spectral reflectance curves and generated spectral reflectance curves for 6 classes in Dataset 0926 are compared in Fig. 22.

The whole graph classification results of the 0928 MS dataset are shown in Fig. 23. In comparison to the direct MS classification and the PSJN classification, the MTSRN classification result is more accurate in classifying the Yellow River water in the middle of the image, which is closer to the HS classification result of 0926 in Fig. 21(h), indicating

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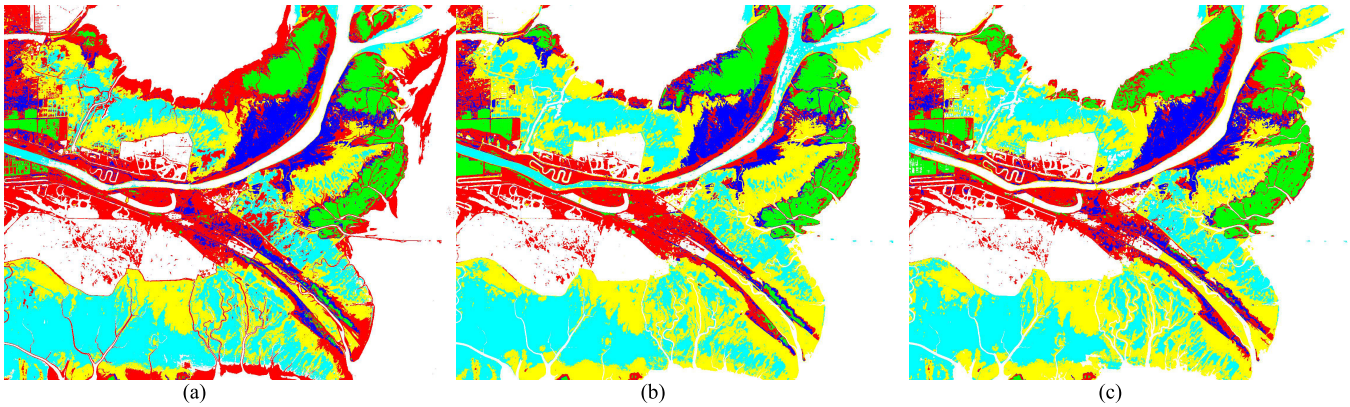


Fig. 23. Whole graph classification result of 0928. (a) Classification result of 0928 MSI. (b) Classification result of 0928 SR HSI by PSJN. (c) Classification result of 0628 SR HSI by MTSRN.

a more precise distinction of spectral features. The coastal areas of spartina also distinguish spartina from reeds more reasonably. The area to the right of the center of the image is also classified as mudflat more in line with ground truth than as reed.

V. Conclusion

This article proposed a MTSRN to reconstruct HS images from short-term continuous MS images. MTSRN consists of two parts: STSRNs and MTFN. STSRN is a reconstruction network that can independently reconstruct MS data of a single-phase into HS data, while MTFN can improve the reconstruction effect by combining neighboring phase information. The MTFN can be switched to single-phase mode or multiphase mode with multiple switches, where the inputs of other phases will not be used in single-phase mode. A post-processing method named PA was applied to enhance the physical plausibility of the reconstructed spectra. Several sets of experiments demonstrate the improvement in SR performance of the MTSRN using multitemporal information compared to the reconstruction network using only single-temporal information. Although the proposed MTSRN has already had excellent SR capability, it can be seen that there is still room for progress in SR across scenes and classes. In future work, we would like to develop SR approaches on multisensor information to improve the models' SR capability further.

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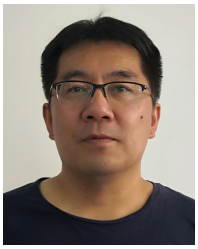
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